



# OPENFLOW TECHNICAL DEEP DIVE

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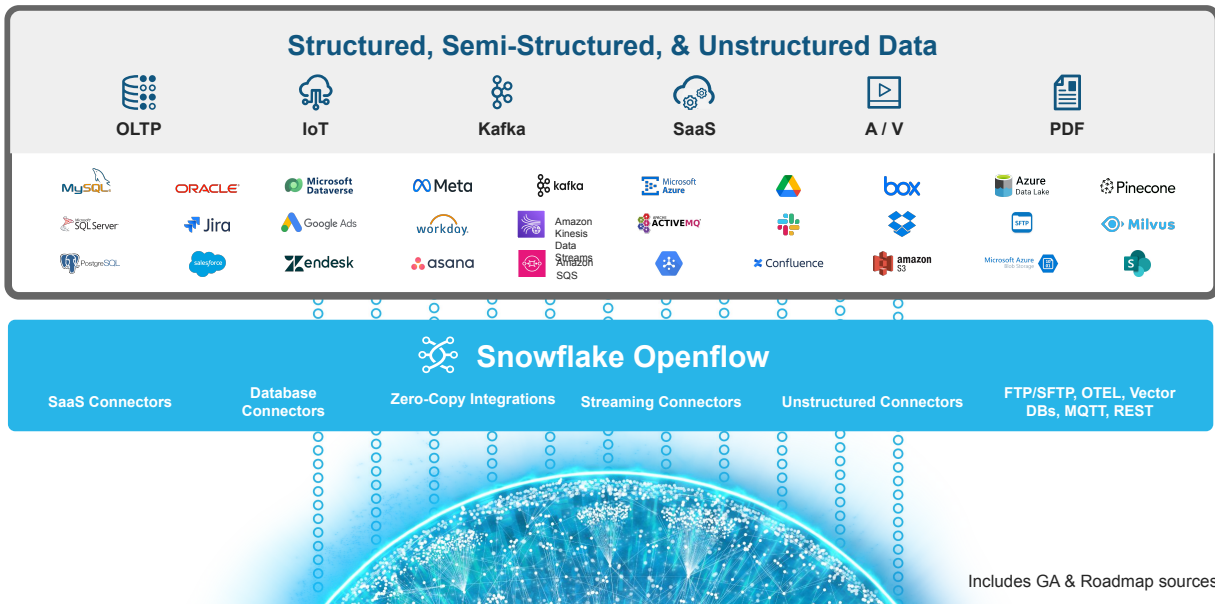


# OPENFLOW OVERVIEW



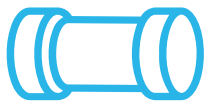
# Snowflake Openflow

- Access **all** enterprise data. Every system of record becomes part of your AI corpus, enabling complete cognitive coverage for agents.
- **Deployment Flexibility** with both Snowflake Hosted and BYOC options so that you securely connect systems even if they are behind your firewalls
- **Hands-free scaling** that eliminates operational overhead





# Ingestion Options for Every Use Case



## Snowpipe

1. Automated ingestion of files from cloud storage
2. Serverless
3. Median latency ~30s
4. Simple, data volume based pricing



## Snowpipe Streaming

1. Near real-time ingestion of rowsets
2. Serverless
3. < 5s median latency
4. Client application needed



## Openflow

1. Enterprise Wide Integrations
2. BYOC & Snowflake hosted options
3. CDC Replication, Streaming, SaaS, Unstructured Data

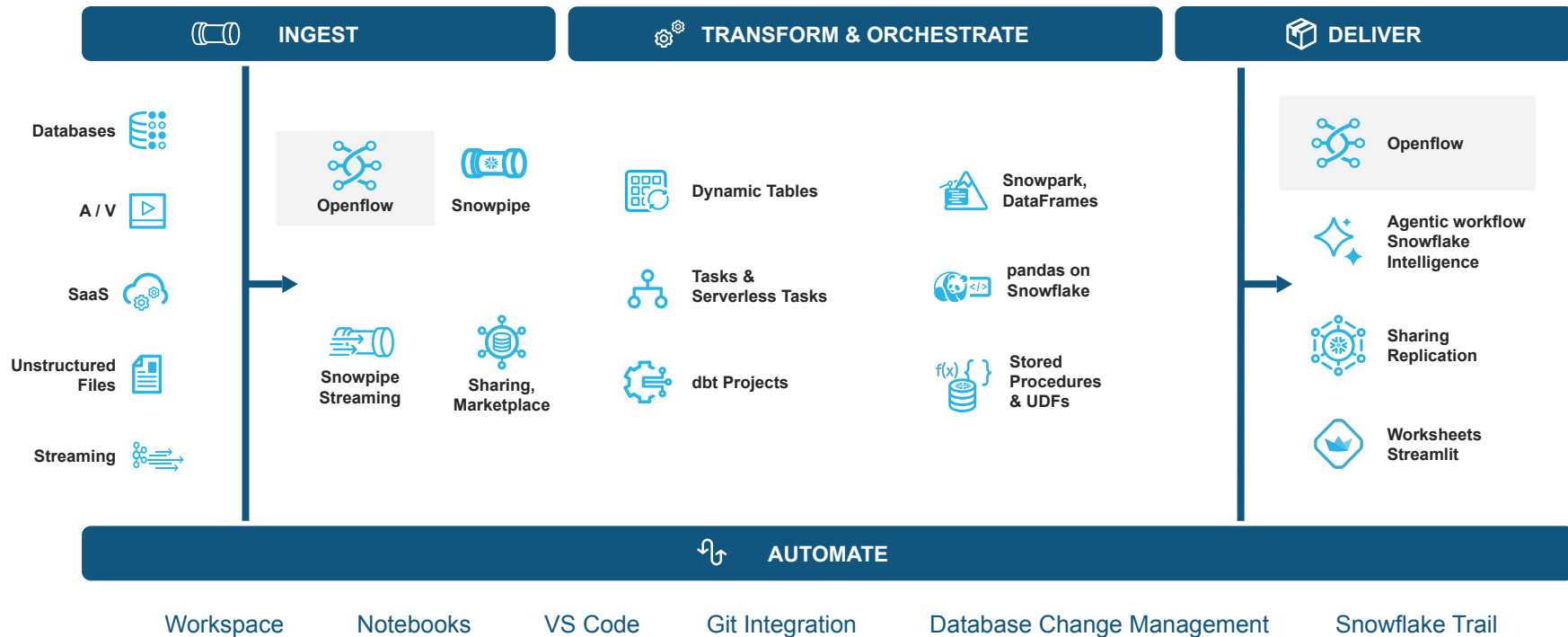


## ETL Partners

1. Leverage existing ETL tooling investment & knowledge
2. Extensive Pre-built Connector Requirements



# Snowflake for Data Engineering



# OPENFLOW CONNECTORS & FLOWS



# NiFi Processor vs Openflow Connector

## Terminology

### NiFi Processor

- NiFi has a web-based user interface for design, control, feedback, and monitoring of dataflows.
- A Processor is like a building block in a Flow
- Link multiple Processors together to build a Flow
- Developed by both Apache and Snowflake



276  
[Processors](#)



88  
[Processors](#)

### Openflow Connector

- Openflow Connectors are curated, versioned Apache NiFi Flow definitions built using open-source and proprietary NiFi components
- Openflow Connectors contain multiple Processors configured specifically for operations with Openflow
- Connectors follow a strict set of design patterns to ensure performance, fault-tolerance, and ease of configuration
- Developed by Snowflake

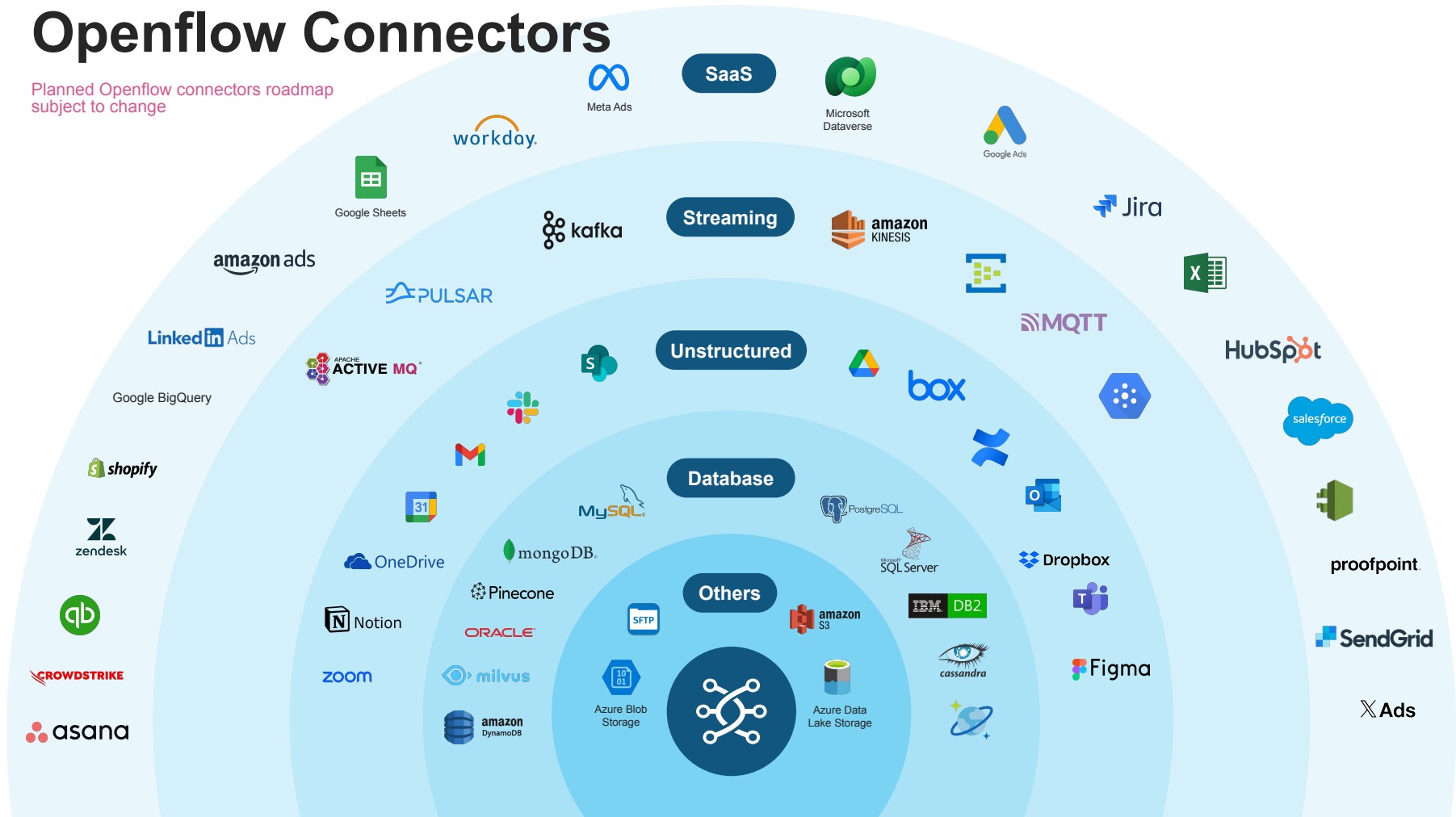


18 (and growing)  
[Connectors](#)



# Openflow Connectors

Planned Openflow connectors roadmap  
subject to change



# Getting Started with Connectors

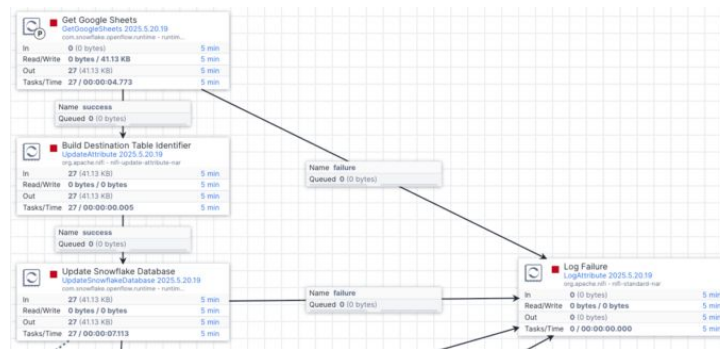
1. Discover and Install a Connector straight from Snowsight.
2. Configure the Source and Destination parameters.
3. Start the Process Group to begin data ingestion/replication.
4. Customize the Flow if needed.

 **Google Sheets** Preview

By Snowflake

Load data from Google Sheets for collaborative reporting and analytics

[Add to runtime](#) [View documentation](#)



# Transformations

## Inflight Transformations



- Event level operations
- Schema / Data Type changes
- Filtering & Replacing
- PII Handling (Masking & Redaction)
- Routing to different tables based on attributes
- Enriching (via lookup, API calls, etc.)
- Data Manipulation (splitting records, joining streams)
- Compressing/Decompressing
- Encrypting/Decrypting

## Stateful Transformations



Dynamic Tables



Snowpark



Tasks



PROCs



dbt Projects

- Operations across events
- Aggregations
- Joins
- Window processing



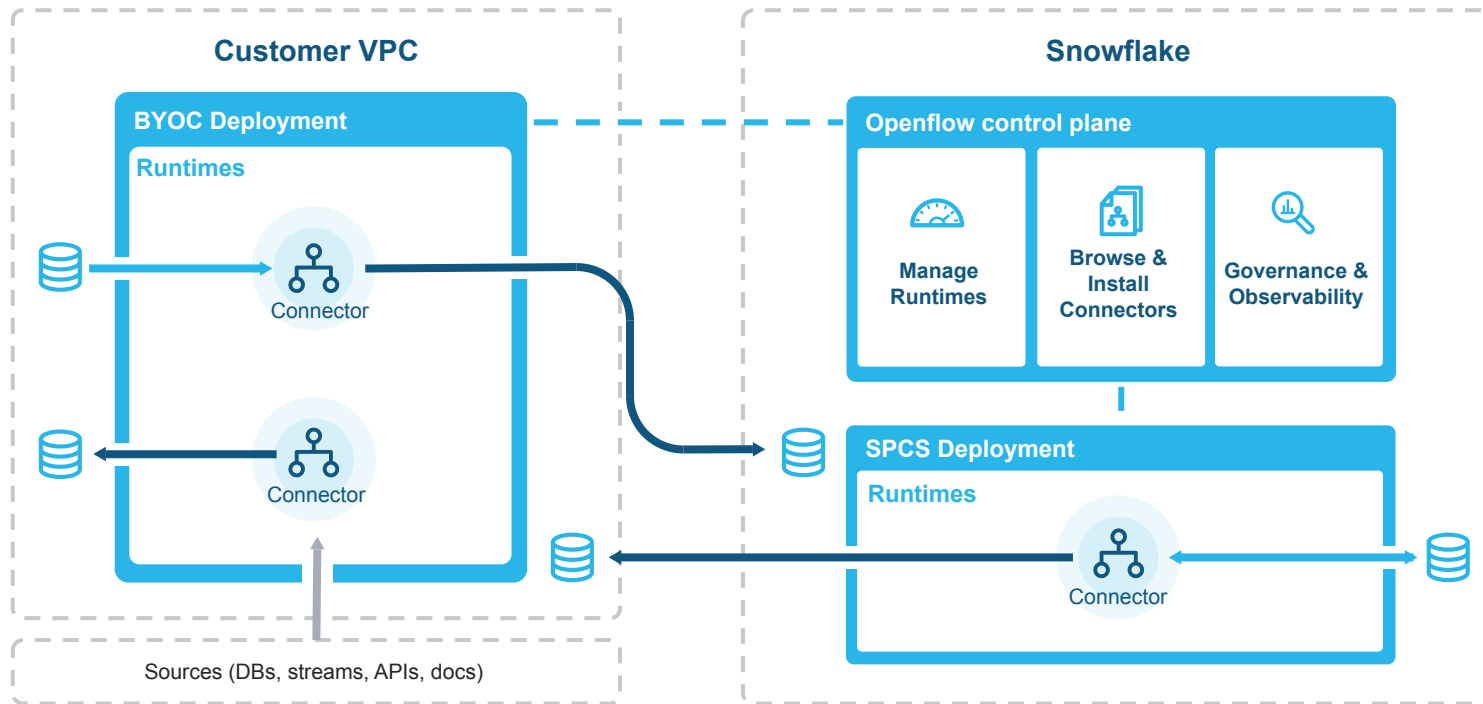
# OPENFLOW ARCHITECTURE





# Handling Global Data Movement

Options across Snowflake and non-Snowflake environments



# Deployment Options



## Snowflake Hosted (SPCS)

- Snowflake Hosted
- Snowflake Managed
- Rapid deployment and immediate access
- No separate environment needed
- Easy data connectivity to public endpoints  
Ex - Workday, SFDC, SharePoint, etc...
- Status
  - AWS: GA
  - Azure: GA
  - GCP: PuPr

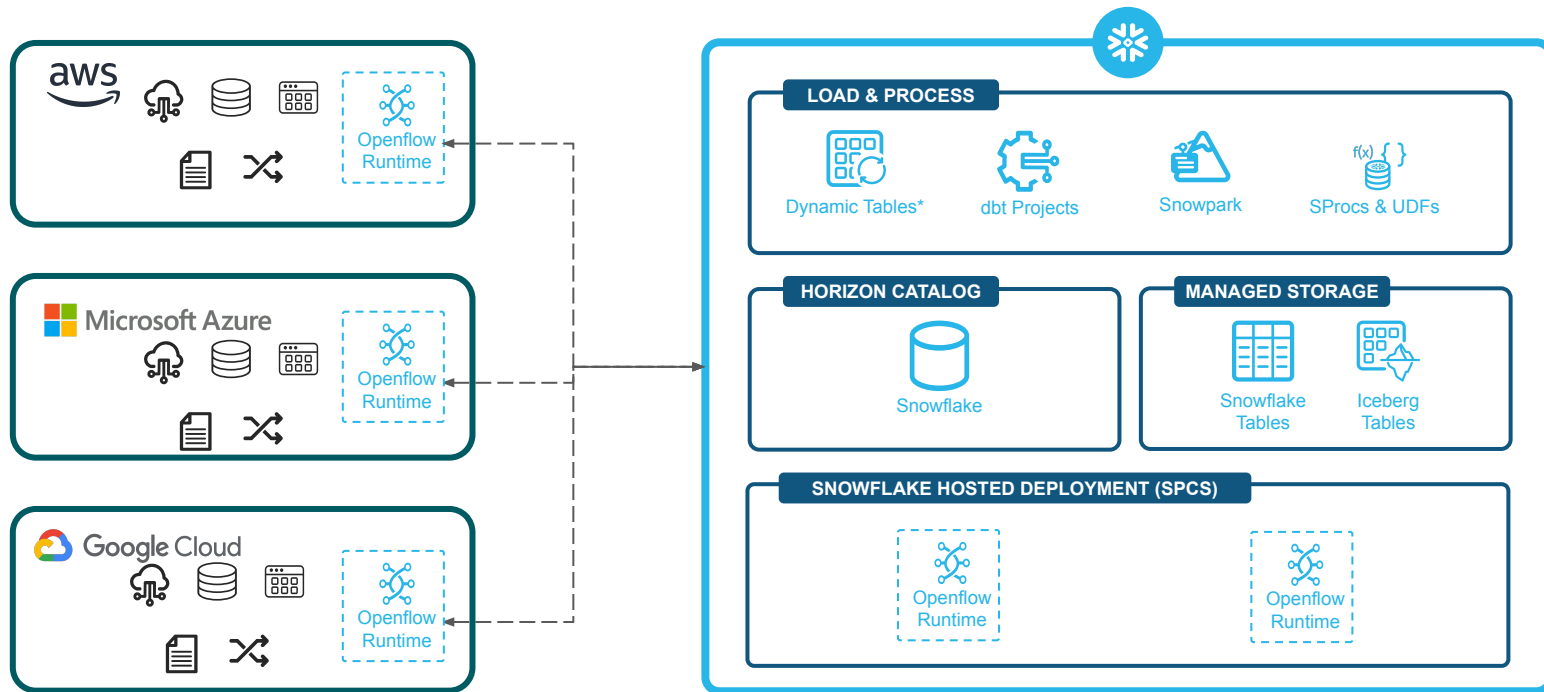


## Customer Hosted (BYOC)

- Customer Hosted
- Snowflake & Customer Managed
- Greater customizations for private networking topologies
- Leverage existing cloud provider pricing
- Pre-process data locally to adhere to data privacy regulations
- Status
  - AWS: GA
  - Azure: GA
  - GCP: PuPr



# Multi Cloud Data Pipelines



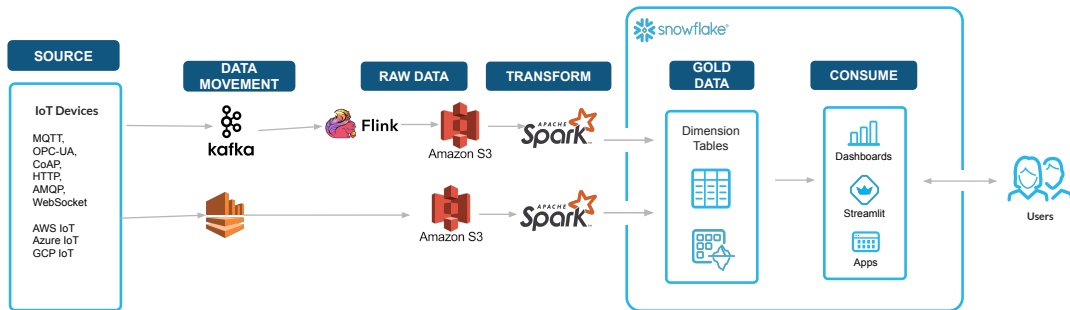
# OPENFLOW USE CASES



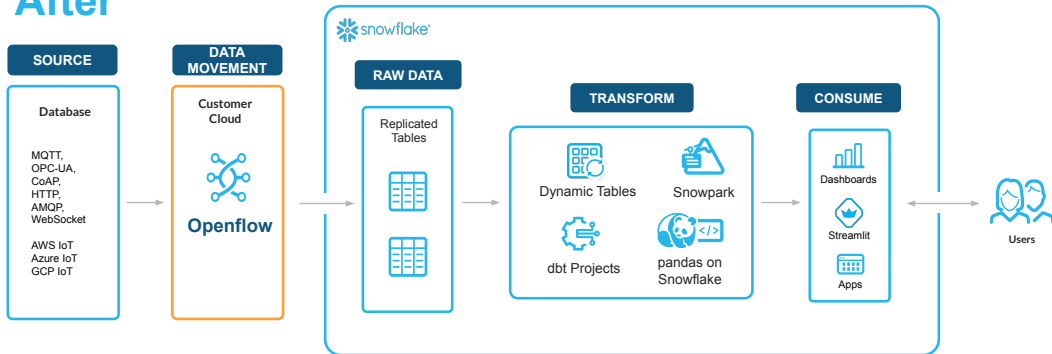
# IoT Flow

## Real-time Data Capture From IoT Devices

### Before



### After



❌ Complex and costly tools that require specialized skills and long set up cycles

❌ Risk to business operations from replication delays or failures

❌ Maintenance headaches with ongoing updates, schema changes and manual troubleshooting

✅ Cost-effective with scalable deployment and reduced TCO

✅ No impact to business operations with lightweight replication

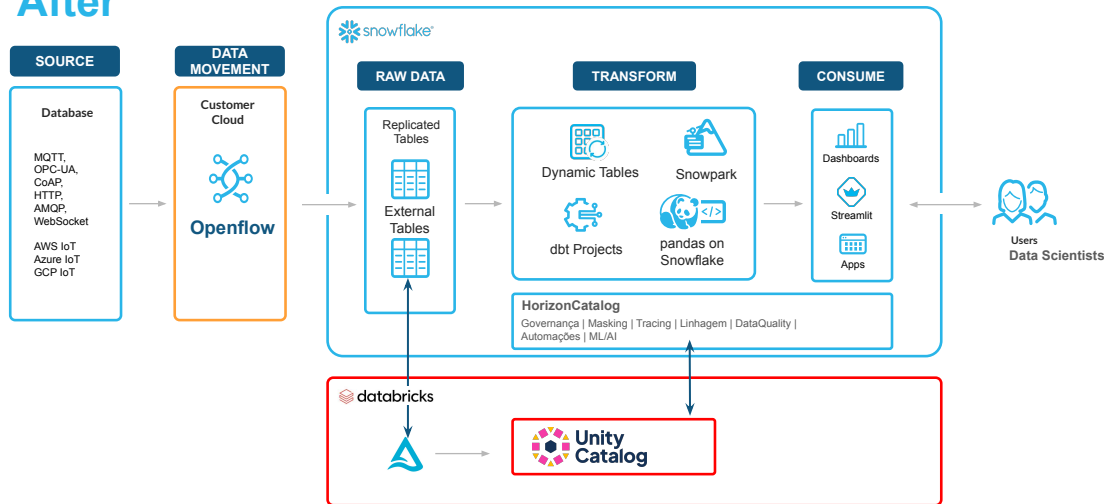
✅ Access near real-time insights with ease using Dynamic Tables



# IoT Flow

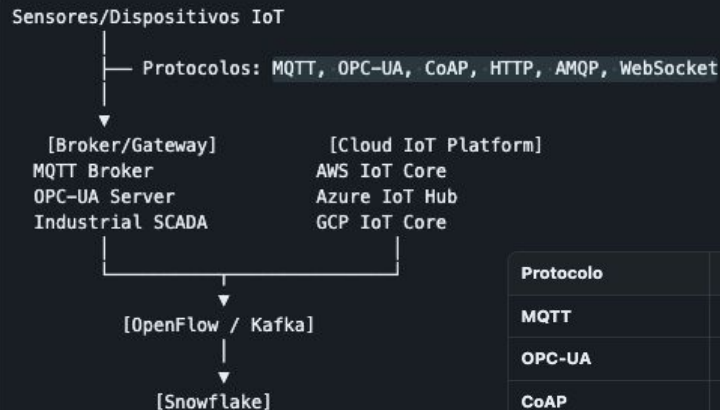
## Real-time Data Capture From IoT Devices

After



# IoT Flow - OpenFlow

## Real-time Data Capture From IoT Devices



| Protocolo     | Processor NiFi/OpenFlow       | Uso Típico                                       |
|---------------|-------------------------------|--------------------------------------------------|
| MQTT          | ConsumeMQTT / PublishMQTT     | Sensores genéricos, Smart Home, telemetria leve  |
| OPC-UA        | GetOPCData / SubscribeOPCData | Industrial IoT, SCADA, manufatura, automação     |
| CoAP          | Processors customizados       | Dispositivos com bateria, baixa largura de banda |
| Kafka         | ConsumeKafka                  | Plataformas de streaming de alto volume          |
| HTTP/REST     | ListenHTTP / InvokeHTTP       | APIs de dispositivos, webhooks                   |
| AWS IoT Core  | Via S3 ou Kinesis rules       | Ecossistema AWS                                  |
| Azure IoT Hub | Via Event Hub connector       | Ecossistema Azure                                |



# IoT Flow - Other Ways

## Real-time Data Capture From IoT Devices

| Método                          | Latência | Melhor Para                                            |
|---------------------------------|----------|--------------------------------------------------------|
| <b>Snowpipe Streaming (SDK)</b> | Segundos | Alto volume, baixíssima latência, SDK Java/Python      |
| <b>Kafka Connector</b>          | Segundos | Kafka já existente na stack de IoT                     |
| <b>Snowpipe (batch)</b>         | 1-5 min  | Arquivos em S3/Azure/GCS (ex: AWS IoT → S3 → Snowpipe) |
| <b>Dynamic Tables</b>           | Minutos  | Transformação contínua de dados brutos IoT             |

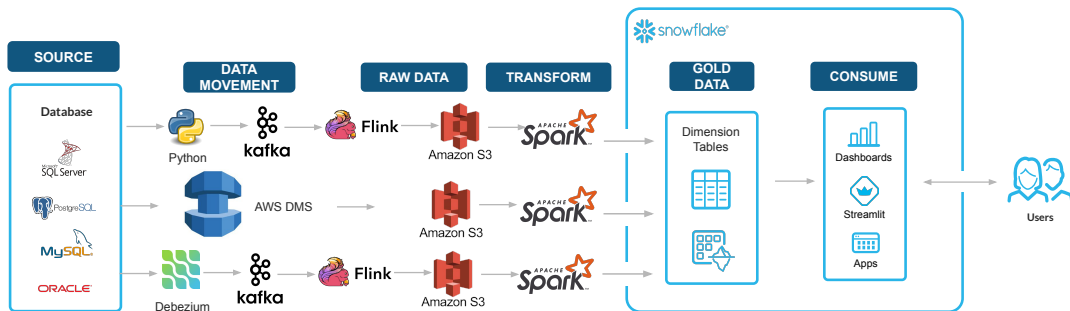




# CDC Replication for Operational Analytics

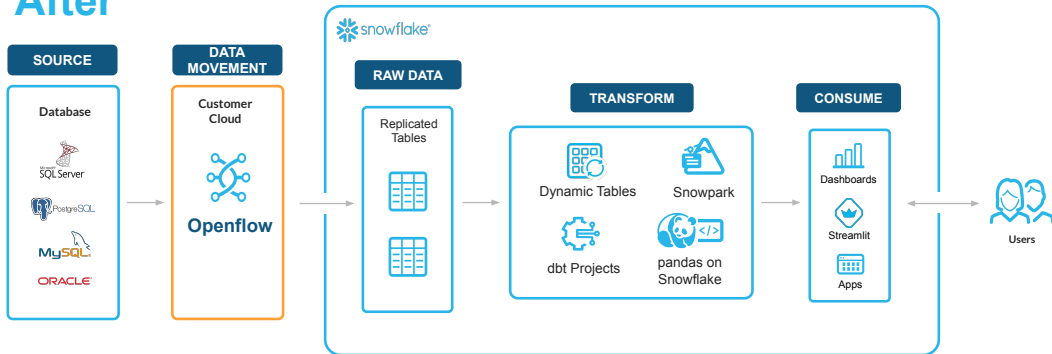
## Near Real-time Data Replication From Transactional Systems

### Before



- ❌ Complex and costly tools that require specialized skills and long set up cycles
- ❌ Risk to business operations from replication delays or failures
- ❌ Maintenance headaches with ongoing updates, schema changes and manual troubleshooting

### After



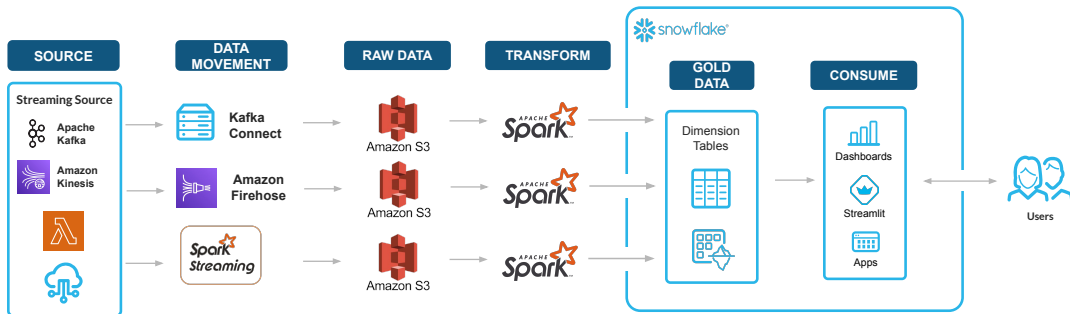
- ✅ Cost-effective with scalable deployment and reduced TCO
- ✅ No impact to business operations with lightweight replication
- ✅ Access near real-time insights with ease using Dynamic Tables



# Near Real-Time Analytics on Streaming Data

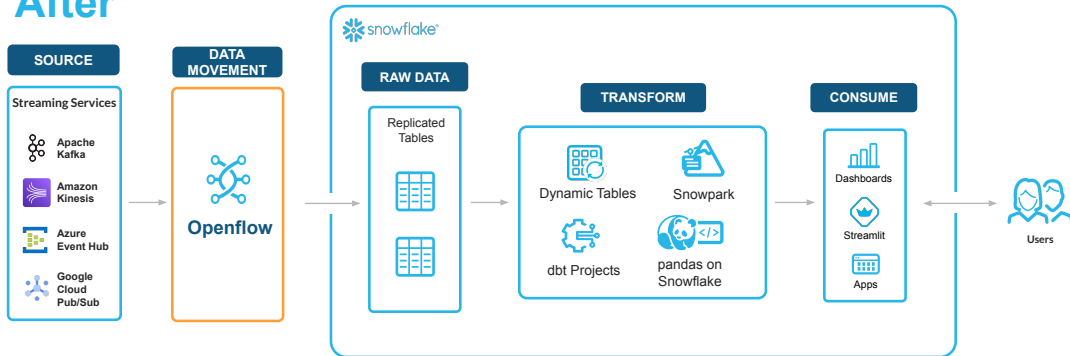
Near Real-Time Insights for Low Latency Use Cases

## Before



- ❌ Too many different streaming services
- ❌ High learning curve across fragmented tools, depends on specialized expertise
- ❌ Limited scalability leads to operational complexity and high costs

## After



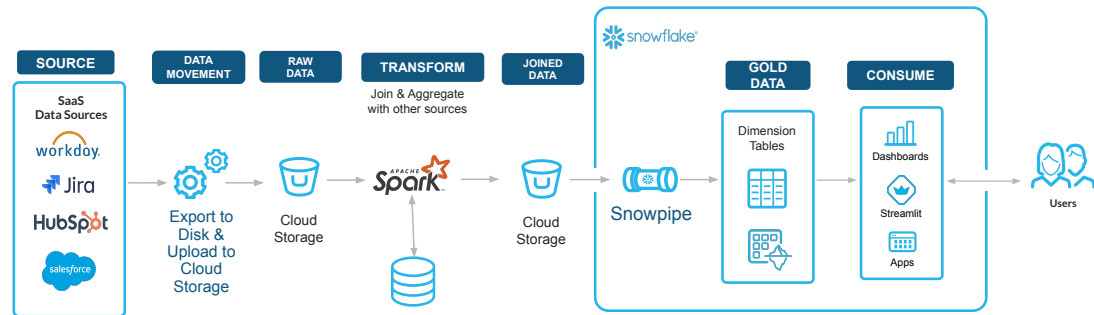
- ✅ A single unified platform to ingest from all streaming services
- ✅ Zero infrastructure management with hands-free scaling experience
- ✅ Cost-effective at scale through optimized resources usage



# Building the Real-Time Customer 360

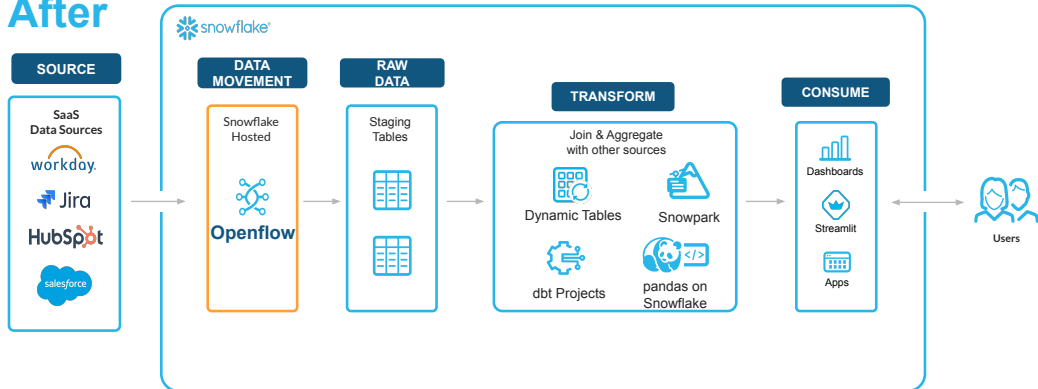
## Holistic Analytics over All Your Data

### Before



- ❌ Fragmented view of customers with disconnected systems
- ❌ Data silos that block teams from accessing the full content
- ❌ Budget unpredictability and cost overruns

### After



- ✅ Unified and transparent platform integrating data from multiple sources
- ✅ Fully understand the customer journey with cross-channel analytics
- ✅ Own and control all the data



# Openflow Observability



# Get Alerts for common Openflow Issues

## Recommended Alerts for Openflow

### WHAT IS IT

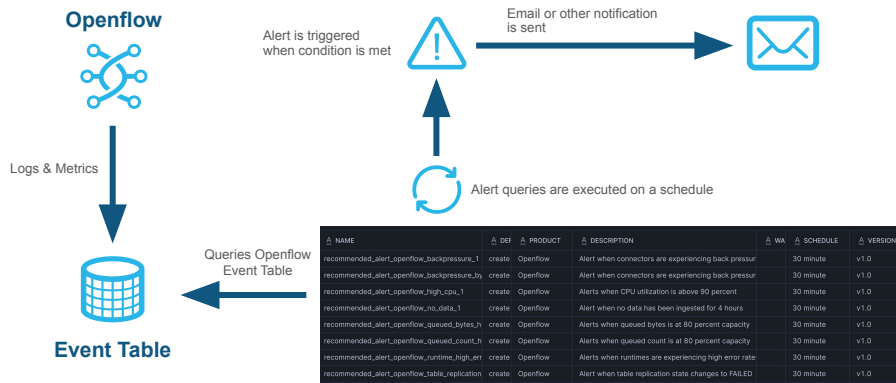
A set of out of the box alert configurations tailored for Openflow. The alerts run on a schedule, query logs and metrics in the event table and fire when conditions are met.

### WHY USE IT

Get alerted when your Openflow connectors are having issues or are not meeting performance expectations. Eliminates the need to reactively query the event table manually or monitoring the Openflow canvas.

### HOW TO USE IT

Create the set of Openflow recommended alerts, configure a notification integration as desired and monitor alert executions via `account_usage.alert_history`



```
CALL SNOWFLAKE.ALERT.CREATE_RECOMMENDED_ALERTS ({
  'openflow': { 'objects_config': [{}]} ,
  'notification_integration': 'my_email_int'
});

ALTER ALERT recommended_alert_openflow_no_data_1
resume;
```



# Monitor Connectors and Troubleshoot Issues (1/2)

## Connector Monitoring Dashboard & CDC Replication Status

### WHAT IS IT

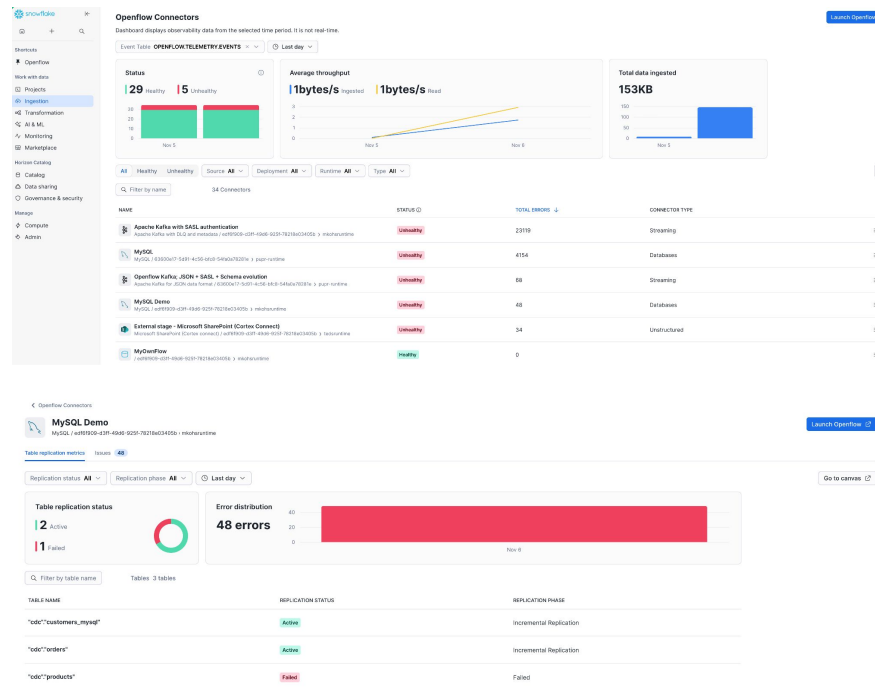
A new experience in Snowsight providing insight into all deployed Openflow connectors, their health and their performance based on telemetry data in the event table.

### WHY USE IT

Understand connector health and performance at a glance without having to enter the Openflow canvas or query the event table. Understand **CDC table replication status**. Debug issues and identify how to fix them.

### HOW TO USE IT

Deploy a connector in Openflow and navigate to Ingestion→Openflow in Snowsight to explore health, performance and debug issues.



# Monitor Connectors and Troubleshoot Issues (2/2)

## Connector Issues & Log Explorer

### WHAT IS IT

A simpler way to debug Openflow connector issues. The 'issues' tab presents errors in an easy to read way. Deep linking into the Log Explorer makes detailed log analysis easy.

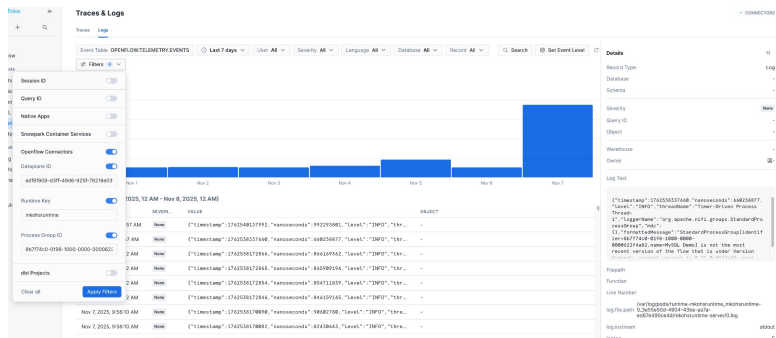
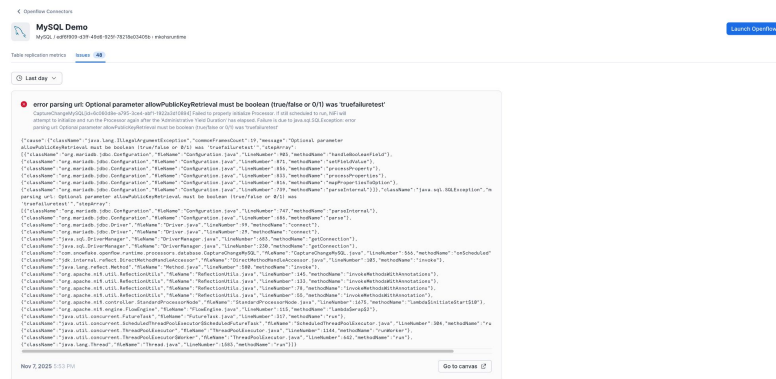
### WHY USE IT

Debug connector related issues and use pre-set filters in the log explorer to find relevant log messages faster.

### HOW TO USE IT

In the Openflow Connectors dashboard open details for a connector and navigate to the 'Issues' tab.

In log explorer, use the new Openflow filters or select 'view logs' for a connector in the dashboard.



# THANK YOU



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